

SEC 2-FA2 GROUP 24-VALDEZ, CJ

<https://github.com/ChanceValdez/APM1110/tree/main/FA2>

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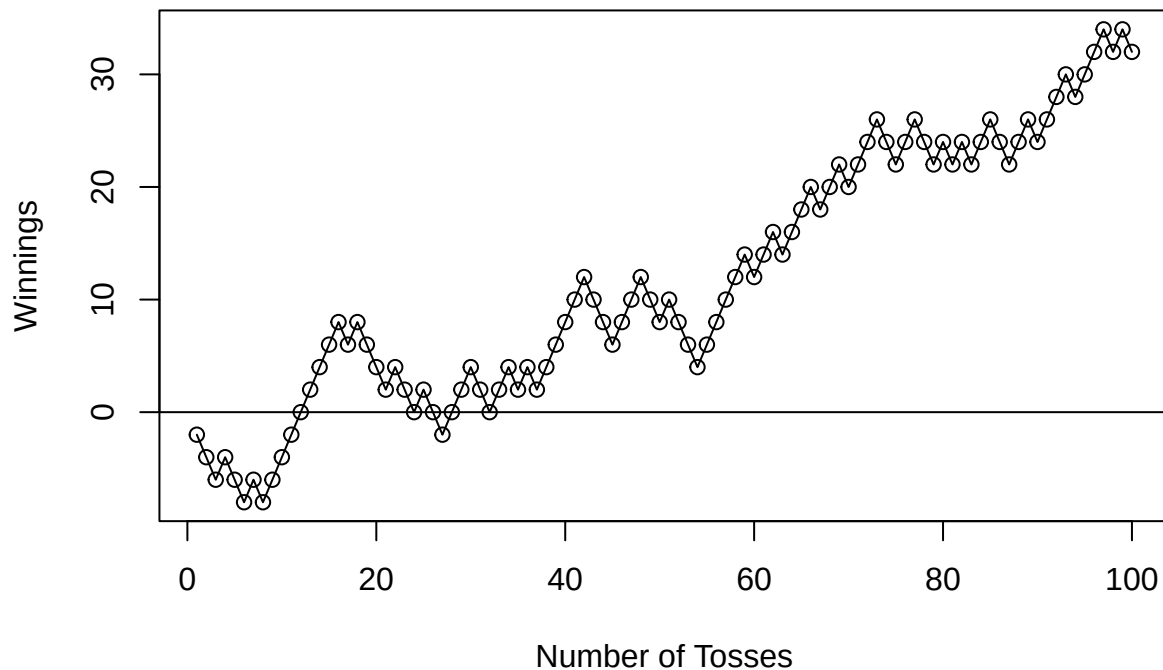
Item 4

Amy and Jane are gambling together. A fair coin is tossed repeatedly. Each time a head comes up, Amy wins two euro from Jane, and each time a tail comes up, Amy loses two euro to Jane. Use R to simulate this game 100 times:

R Code to Simulate Tossing Game

```
num_tosses <- 100
H <- 2
T <- -2

tosses <- sample(c(H, T), size = num_tosses, replace = TRUE)
winnings <- cumsum(tosses)
plot(1:num_tosses, winnings, type = "o",
     xlab = "Number of Tosses", ylab = "Winnings")
abline(h = 0)
```



Now, estimate:

- (a) the number of times that Amy is ahead in these 100 tosses;
- (b) how much Amy has won or lost after 100 tosses.

R Code to Answer (a) and (b)

```
times_ahead <- sum(winnings > 0)
cat("The number of times Amy was ahead in these", num_tosses, "tosses:\n")
```

```
## The number of times Amy was ahead in these 100 tosses:
```

```
cat("Amy was ahead:", times_ahead, "times.\n")
```

```
## Amy was ahead: 83 times.
```

```
final_winnings <- winnings[num_tosses]
cat("How much Amy has won or lost after", num_tosses, "tosses:\n")
```

```
## How much Amy has won or lost after 100 tosses:
```

```
cat("Amy's final net winnings/losses:", final_winnings, "euros.\n")
```

```
## Amy's final net winnings/losses: 32 euros.
```

Additional Calculations (for fun!)

```
times_tied <- sum(winnings == 0)  
times_behind <- sum(winnings < 0)
```

```
cat("Amy was tied (at 0 euros):", times_tied, "times.\n")
```

```
## Amy was tied (at 0 euros): 5 times.
```

```
cat("Amy was behind:", times_behind, "times.\n")
```

```
## Amy was behind: 12 times.
```

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